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Rediscovering Relativity Is Not a Single Task
What It Takes for AI to Discover a Physical Theory

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Can an AI system rediscover general relativity? This question is increasingly discussed in the AI community, but it is often framed too narrowly. Some accounts implicitly assume that rediscovery means reproducing Einstein's field equations; others assume that a single batch of experimental data suffices to construct a theory; still others reduce scientific discovery to one mode of inference. We argue that rediscovering a physical theory is not a single task, but a layered sequence of distinct capabilities. Using the history of relativity as a case study, we reconstruct it not as a linear path from Newton to Einstein but as a branching theory space populated by Lorentzian, Poincaréan, scalar, vector, and geometric theories. We show that theory construction and experiment design were mutually generative rather than sequential, and that at several points the available evidence could not adjudicate among live candidates. On this basis we propose a decomposition of rediscovery into distinct capabilities, and three staged tests for AI systems in theoretical physics.

1. Introduction

Can AI rediscover general relativity? This question has recently become a benchmark for evaluating the scientific capabilities of AI systems (Wang et al., 2023). One influential proposal is Hassabis's "Einstein test": train a system whose knowledge is restricted to what was known in 1911 and ask whether it can independently arrive at general relativity, just as Einstein did in 1915. Only then, the argument goes, could we claim that the system possesses genuine general intelligence.[1] One such attempt trains models from scratch on corpora containing only pre-1900 texts, aggressively removing any traces of later concepts (Hla, 2026).

The motivation behind these efforts is sound: they try to hide the answer and reduce data contamination. But they still leave the central question underspecified. What counts as "rediscovering" relativity, and how should success or failure be evaluated along the way? This is the gap this paper addresses. In our view, rediscovering relativity is not about recovering a single answer; it is about reconstructing a historical theory space.

So far, the notion of "rediscovery" itself is ambiguous. Does it mean reproducing Einstein's field equation? Does it mean reconstructing the entire logical path from special relativity to general relativity? Or is reproducing only one crucial step sufficient? Different answers correspond to fundamentally different capabilities, yet current discussions often arrive at yes-or-no conclusions without first distinguishing among them.

More importantly, the current discussion often makes the history of relativity look too static, as if all relevant facts had already been collected and the task were to infer the right theory from a fixed archive. The actual history was not shaped in this way. Maxwell's equations predicted electromagnetic waves and incorporated light into electromagnetism, which in turn motivated the search for the so-called "aether wind". The null result of the Michelson–Morley experiment led to Lorentz's contraction hypothesis. The experiments carried out between 1902 and 1904 were no longer testing for the aether wind, but the new predictions generated by the contraction hypothesis itself. The history of relativity was an interaction between theory and experiment: theories suggested what to test next, and experiments forced theories to change. From Maxwell to Lorentz, from Poincaré to Einstein, and later to the competing theories of relativistic gravity, the development of relativity was a continuous process of mutual interaction rather than a one-shot inference.

Instead of discussing rediscovery as one big question, we turn it into three more concrete discovery questions, each closer to the physics and to the historical details:

- Can AI construct and revise theories as new experimental evidence arrives, as Lorentz did?
- Can AI elevate a collection of empirical regularities into a unifying principle, as Poincaré did?
- Can AI construct the entire space of candidate relativistic theories of gravity, derive their predictions, and

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[1] Demis Hassabis, remarks in "Fireside Chat: Sir Demis Hassabis and Prof Govindan Rangarajan," Indian Institute of Science, 17 February 2026, moderated by Varun Mayya. Transcript available at the IISc EECS event page.

identify the experiments that could distinguish among them?

These are not three versions of the same capability. Together, they ask whether a system can move through a theory space, revise that space as experiments arrive, and identify what must be tested next. The boundary is equally important: theory can tell us what would be informative, but not what observation will show.

Our position is that rediscovering relativity requires reconstructing a historical theory space, repeated interactions between theory and experiment, and movement across different modes of reasoning. It cannot be reduced to induction, deduction, or a single act of abduction, nor can it be completed entirely on paper.

Section 2 reconstructs this theory space. Section 3 reviews three common ways of framing the problem. Section 4 decomposes rediscovery into distinct capabilities. Section 5 proposes three staged evaluations. Section B contrasts the position of a system doing mathematics with that of a system doing physics.

2. The Historical Theory Space of Relativity

2.1. 19th-Century Situations

Newtonian mechanics and universal gravitation had won enormous authority in 19th-century astronomy. The standard example was Neptune: irregularities in the motion of Uranus were used to infer the position of an unseen planet, which was then observed. Mercury's anomalous perihelion precession was a problem, but not yet a reason to abandon Newtonian gravity. The residual was about $43''$ per century, out of a total precession of roughly $5557''$.

The first responses were patches inside the Newtonian picture. Le Verrier proposed an intra-Mercurial planet, Vulcan, in 1859 (Le Verrier, 1859), and astronomers searched for it for decades without confirmation. Other fixes included solar oblateness and Hall's 1894 modification of the inverse-square exponent (Hall, 1894). These proposals had weaknesses, but Mercury alone did not force Einstein's answer or open up new physics. As long as old models still had plausible adjustments, a patched Newtonian theory remained the natural first move.

Electrodynamics created a different situation. Maxwell's equations unified electricity, magnetism, and light by treating light as an electromagnetic wave. But the equations were not covariant under Galilean transformations, which made the aether look like the natural rest frame for electromagnetic waves. The Michelson–Morley experiment (Michelson & Morley, 1887) was designed to detect the aether wind, the motion of the Earth through this frame. Its result was null. By the end of the 19th century, relativity was not born from one clean anomaly. It came out of two live situations: a small gravitational residual inside a very successful Newtonian programme, and an electrodynamic theory whose own structure made the aether experimentally suspect.

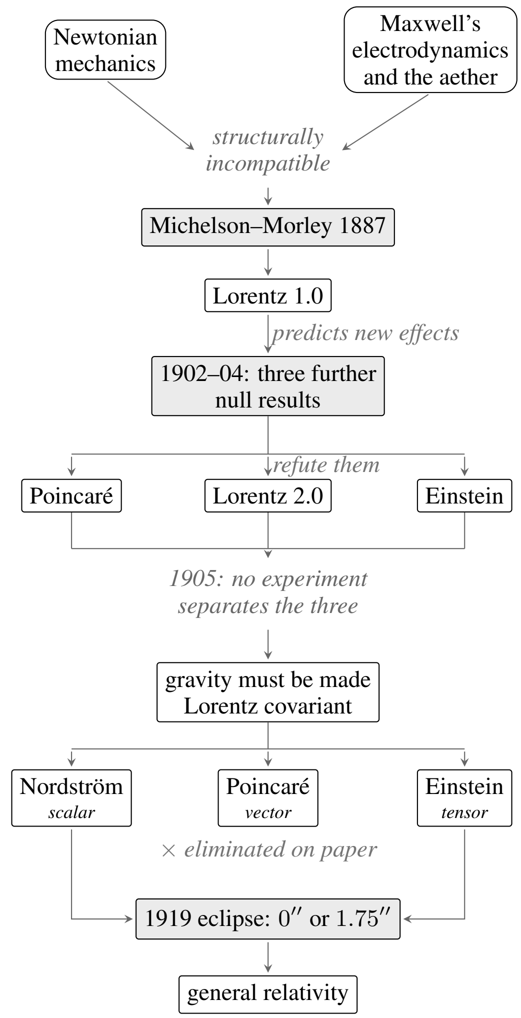

Figure 1. The theory space, from the two sources to general relativity. Shaded boxes are the points at which the world had to answer, the same convention used in Tables 1 and 2. The vector theory is eliminated before any observable is computed. The scalar theory survives every calculation and is eliminated only in 1919.

2.2. Special Relativity: Lorentz, Poincaré, Einstein

Lorentz's theory lived inside the aether framework. His 1895 theory (Lorentz, 1895) used local time and length contraction to explain why the Michelson–Morley experiment did not detect aether drift. Later Rayleigh, Brace, and Morley–Miller tested consequences of the FitzGerald–Lorentz contraction idea (FitzGerald, 1889); Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque. Lorentz had to update his theory with a more elaborate electron model and additional stabilising assumptions to match these new experiments (Lorentz, 1904), but he did not abandon the aether. In this sense, Lorentz was still patching a physical picture.

Compared with Lorentz's increasingly complex theory, Poincaré started from a simpler first principle (he called it "the principle of relativity"). In 1904, he treated the impossibility of detecting aether drift as a general postulate (Poincaré, 1904). He corrected Lorentz's formulation, showed that the Lorentz transformations form a group (Poincaré, 1906), and made the symmetry requirement explicit. Physically, Poincaré still accepted the aether framework and treated the time and length measured in other frames as artificial quantities. Mathematically, this brought his theory very close to Einstein's.

Unlike the other two, Einstein removed the privilege of the aether and the absolute frame. His 1905 work (Einstein, 1905) started from the relativity principle and the constancy of the speed of light, without referring to any hidden rest frame. The time t' of a moving inertial frame was not a calculational fiction, as Poincaré suggested, but the real measurable time of that frame. Today this is the physical interpretation we usually identify with special relativity. But in 1905 there was no experiment that could separate Lorentz, Poincaré, and Einstein by their observable predictions. For the experiments at issue, Lorentz's, Poincaré's, and Einstein's theories gave the same observable results, even though they rested on different physical pictures.

2.3. Theory and Experiment as Mutually Generative

We clarify the interaction between theory and experiment. Between 1887 and 1902, there was only one aether-drift experiment - the Michelson–Morley experiment. Lorentz's 1895 theory of length and local time was good enough to explain the Michelson–Morley result. However, the next experiments were different. Rayleigh and Brace tested whether FitzGerald–Lorentz contraction should produce double refraction in moving matter (Rayleigh, 1902; Brace, 1904). Morley and Miller tested whether the effect depended on the material construction of the interferometer (Morley & Miller, 1905). Trouton–Noble tested whether a charged capacitor moving through the aether should feel a torque (Trouton & Noble, 1903). These were not just more searches for aether wind. They were new questions created by the theory itself.

All of them gave null results. This is a typical example of theory and experiment moving together: Lorentz's theory explained one experiment, then inspired new experiments, and those experiments pushed him from contraction alone toward the full Lorentz transformation and a more complicated electron theory. After that, no new decisive experiment arrived. Poincaré and Einstein then rebuilt the same experimental situation in cleaner ways, each simpler than Lorentz's physical picture, but neither reformulation was selected by a new observation.

2.4. Relativistic Gravity Before General Relativity

Special relativity is naturally described in Minkowski spacetime, but Newtonian gravity does not fit there. Poisson's equation is not Lorentz covariant. How should gravity be brought into Minkowski spacetime? Between 1905 and 1915 this was an open problem worked on by Poincaré, Nordström, Abraham and Mie, not by Einstein alone (Walter, 2007). The standard move was to keep spacetime flat and add a gravitational field on it, exactly as one adds an electromagnetic field. The field has to be Lorentz covariant, it has to reduce to Newtonian gravity in the static weak-field limit, and its source has to be the energy-momentum tensor $T_{\mu\nu}$.

Poincaré took the vector in 1905, modelling gravity on electromagnetism (Poincaré, 1906; Walter, 2007), where the source is a four-current j^μ and the field a four-potential A^μ. It was the only template available, since the energy-momentum tensor did not yet exist as a tool (Minkowski, 1908; von Laue, 1911). It gave gravitation a propagation speed c and what Poincaré called gravitational waves, but little else to test with observation: a perihelion advance for Mercury well short of the observed $43''$, and no other prediction.

Nordström kept Newton's potential as a scalar and kept it as a field on Minkowski spacetime (Nordström, 1913; Norton, 1992). Newton's equation contains derivatives in space only, which is why its gravity acts everywhere at once. Nordström added the missing time derivative, which turns it into a wave equation and makes the potential travel at c, and he replaced mass density as the source by the energy of matter, since in relativity the two are the same thing. His theory was self-consistent and attractive, and it explicitly requires $m_i = m_g$, so the equivalence principle was not Einstein's private concern, and the gravitational redshift comes out right for that reason. However, he calculated a perihelion precession of $-7''$ per century for Mercury, with the sign reversed, and no bending of light passing through the gravitational field of matter.

Einstein started on the same flat background, and with the same choice as Nordström. From 1907 to 1912 he worked with a scalar, using a position-dependent speed of light $c(x)$, and the $0.83''$ deflection he published in 1911 (Einstein, 1911) came from it, half of the value now measured. He gave the scalar up when it could not be carried beyond static fields or made to hold the full equivalence principle.

What he turned to was not another field placed on Minkowski spacetime. He took gravity to be the geometry of spacetime itself, and that settles the rank without any further choice: geometry is carried by the metric in $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$, and the metric is a symmetric second-rank tensor by construction. The rank was not selected. It was handed over by the geometry.

Having the idea was not having the theory. The field equations took three more years and arrived in November 1915 (Einstein, 1915a;b), and only then did the predictions come out right: a perihelion precession of $+43''$ per century for Mercury, and a deflection of $1.75''$ for light grazing the Sun, twice the value he had published in 1911.

3. Common Framings of Rediscovery

3.1. The Whig History Framing

Whig history (Butterfield, 1931) writes the past from the "winning end". It keeps the line that leads to the present, drops the branches that left no descendants, and in doing so makes the outcome look like the only thing that could have happened. Applied to relativity, it records Einstein and his reasoning alone, and omits the parallel work of Lorentz, Poincaré, Nordström, Abraham, Mie and so on.

The resulting account is familiar. Special relativity rests on two postulates, the principle of relativity and the constancy of the speed of light. General relativity begins from an anomaly, the $43''$ per century left unexplained in Mercury's perihelion, proceeds through the equivalence principle, takes Riemannian geometry as its mathematical tool, and arrives at Einstein's field equations coupling matter to the geometry of spacetime, which displaces the Newtonian paradigm.

As a framing of the discovery problem, it carries a further assumption: at each historical moment, there was already one definite correct answer, and discovery meant reaching it. This assumption makes the framing operational. An AI evaluation built on it can score a system by asking whether it outputs the two postulates of special relativity, or the Einstein field equations, and count anything else as failure. This framing appears in the Einstein test quoted in Section 1, in benchmark designs that train on pre-1900 corpora and then check for traces of later concepts (Hla, 2026), and in popular accounts that follow Einstein alone.

3.2. The One-Shot Framing

On this view, once the data are in place, the remaining task is to apply the right form of inference and get the theory out. The pipeline is linear: experiments first produce data, the data are then processed, and a theory comes out at the end (Brunton et al., 2016; Udrescu & Tegmark, 2020). Discovery is the processing step. The theory works on an archive that is already in place when the work begins.

Applied to relativity, this account says the following. By the turn of the century, the experiments on the propagation of light were already in hand, and special relativity follows from them. Mercury's anomalous perihelion precession was also already known from nineteenth-century astronomy. Once special relativity is available, that old discrepancy becomes the remaining input from which general relativity follows. Each theory appears in a single step, as the correct reading of the evidence available at the time.

Moved into AI, the framing gives a clean experimental design: give a system all of physics up to 1905 and have it produce general relativity. It appears in public statements by the leadership of AI laboratories, and in benchmark designs that supply a complete corpus in a single prompt and then check whether the target theory is present in the output (Hla, 2026; Lu et al., 2024). It presupposes that the evidence base is fixed before theorising begins, so that all of discovery can be placed in one inference from data to theory.

3.3. The Single-Mechanism Framing

On this view, scientific discovery comes down to a single mode of inference. There is one operation that turns what is already known into a theory. Building a discovery system then means finding that operation and implementing it well. Different accounts name different operations, and each treats the operation it names as the whole task.

Applied to relativity, the episode becomes a case where a single mode of inference worked. In the inductive reading, Einstein found the shortest description covering the observations, and the theory is the compressed form of the data. In the deductive reading, the postulates were already in place and the theory is what follows from them, so the work is derivation. In the abductive reading, the theory arrived in a single leap to the best available explanation, and what came afterwards was elaboration.

Each reading has a current research programme attached to it. The compression line runs through Schmidhuber's treatment of discovery as compression progress (Schmidhuber, 2009). The deductive line runs through formal mathematics and systems such as AlphaProof (Hubert et al., 2025), where results are obtained by searching a fixed set of axioms. The abductive line treats discovery as one explanatory jump and holds in addition that such a jump requires embodied simulation (Zahavy, 2026). What the framing presupposes is that a single mechanism accounts for the whole episode, so that progress on that mechanism is progress on discovery itself.

4. Rediscovery Is Not One Task

Two facts often placed at the origin of relativity did not by themselves produce it. The equality of inertial and gravitational mass had been known for two centuries, but most theories simply had to respect it. It was not yet treated as the problem to solve. Mercury's unexplained $43''$ per century was a real discrepancy, but Newtonian fixes could still absorb it. The new framework came from a different pressure: Newtonian mechanics and Maxwell's electrodynamics did not fit together. Maxwell's equations single out a speed and are not covariant under Galilean transformations, while experiments failed to detect the aether wind that the old picture expected.

That incompatibility opened the problem rather than settling it. What followed took sixty years, and it was not a single step. It included an explanation of the null result still under old framework, revision under a second round of experiments, a new principle, special relativity, the separate problem of making gravity Lorentz covariant, a space of candidate theories, and finally an observation that chose among them. Table 1 sets out the task chain.

Table 1 has one central lesson. Relativity was not a task that could be completed by theory alone. Some rows are experiments: theory can say what should be measured, identify the relevant apparatus, and specify the required precision, but the observed answer has to come from the world. Nor can theory by itself decide which candidate is best. By 1905, Lorentz's, Poincaré's and Einstein's accounts agreed with every experiment then available, and the scalar, vector and tensor theories of gravity were separated only later, by observations not available when the theories were built. The task is therefore a chain: theory proposes, experiment tests, theory revises, and only then does the next theoretical move become possible.

The implication for AI is direct. If an AI system is asked to rediscover relativity, it should not be judged as if the task were to produce the final theory in one step. It can provide the theoretical work: recognise problems, build candidate theories, derive predictions, and design the experiments that would separate them. But the full loop cannot be closed inside the system. At several points, parallel theories fit the evidence already available, and the deciding information had to come from experiment. AI rediscovery should therefore be staged and world-coupled, not an automatic derivation from a fixed archive.

5. Three Staged Tests

The decomposition in Section 4 can be turned into evaluations. We describe three tests, each built from one moment in the chain: Lorentz's revision under new experiments, Poincaré's move from compensations to a principle, and the construction of the space of relativistic theories of gravity. Each test has the same structure. We say what is supplied, what is withheld, what the system is asked to do, and what would count as success.

These are conceptual designs rather than benchmarks ready to run. Every answer is present in any modern training corpus, so a system that produces Einstein's postulates has shown nothing about discovery. Success is uninformative and only failure carries information. What restores diagnostic value is counterfactual variation (Wu et al., 2024): alter the symmetry of the source, or the number of spacetime dimensions, so that the historical answer is the wrong one and the reasoning has to be done rather than recalled. We state the tests in their historical form because that is the form a reader can check.

5.1. The Lorentz Test

The first test is staged, and the staging is the point. The system receives Maxwell's equations, the aether framework, and the null result of Michelson–Morley. The experiments of 1902 to 1904 are withheld, along with the Lorentz transformation, the relativity principle, and everything after them.

It is asked to explain the null result, and then to derive what else that explanation implies. If lengths contract along the direction of motion, what else should be visible, and in what apparatus? Only once those predictions are recorded does the second batch arrive. Rayleigh and Brace tested double refraction in moving matter. Morley and Miller tested whether the effect depends on the material the interferometer is built from. Trouton and Noble tested the torque expected on a moving charged capacitor. All three results are null. The system must then decide whether to patch its hypothesis, abandon it, or rebuild.

Success is not reaching the Lorentz transformation. It is deriving, from the system's own hypothesis, the predictions that the second batch will test. It is also recognising that those predictions have been refuted, and being able to say what the repair has cost in added assumptions. This is the capability listed against tasks 2 and 4 in Table 1. It cannot be assessed at all if the two batches arrive together.

5.2. The Poincaré Test

The second test supplies no new evidence at all. The system receives Maxwell's equations and the whole set of null results at once, and is asked to reorganise what it already has. Withheld are the relativity principle, the Lorentz group and Einstein's postulates. The difficulty here is not revision but elevation: a set of local patches has to be replaced by a single statement.

The failure this test is built to catch is a better patch. A system that returns a more elaborate electron model accounting for every null result has fitted all the data without making the move, since each result is still paid for separately. What is wanted is a prohibition: no experiment can detect uniform motion through the aether. A prohibition also cov-

Table 1. The relativity episode decomposed into tasks. Shaded rows are the tasks that have to be settled in the world. "Design only" means an AI can specify the experiment, down to apparatus and required precision, but a human has to build it and run it, and the answer comes from the world rather than from the reasoning.

#	Task	Capability it requires	Can AI attempt it?
0	Newtonian mechanics and Maxwell's electrodynamics	starting point	—
1	Michelson–Morley: measure the speed of the aether wind	turning a theoretical question into an apparatus	design only
2	Lorentz 1.0: contraction and local time	theory construction on incomplete evidence	yes (the Lorentz test I)
3	Rayleigh, Brace, Morley–Miller, Trouton–Noble	deriving new testable questions from a theory	design only
4	Lorentz 2.0: the full transformation	revision under a second round of null results	yes (the Lorentz test II)
5	Poincaré: the relativity principle	principle formation from accumulated patches	yes (the Poincaré test)
6	Einstein: special relativity	reformulation without a privileged frame	yes (the Poincaré test)
7	Pose the next problem: make gravity Lorentz covariant	problem recognition	yes (the gravity test)
8a	Scalar, vector and tensor candidates	hypothesis-space construction	yes (the gravity test)
8b	Predictions for each candidate	derivation and screening on paper	yes (the gravity test)
9	Deflection of starlight measured at a solar eclipse, 1919	identifying the discriminating observation	design only
10	Gravitational redshift confirmed, 1959	later confirmation, does not discriminate	design only
11	Gravitational waves detected, 2015	later confirmation	design only

Table 2. The three candidates, in order of the rank of the field. The vector is disposed of before any observable is computed, which is why its entries are empty. Shaded rows are the candidates that survive on paper and can only be separated by going to the world.

Rank	Field	Couples to	Mercury precession	Light deflection	Redshift	Outcome
0	scalar φ	$T^\mu_{\ \mu}$, the trace	$-7''$, sign reversed	exactly zero	correct	requires observation
1	vector A_μ	mass current	not computed	not computed	not computed	eliminated on paper
2	tensor $h_{\mu\nu}$	$T_{\mu\nu}$	$+43''$	$1.75''$	correct	requires observation

ers the experiments nobody has performed yet. Success is that principle, the transformations derived from it, and the demonstration that they form a group, closure being the mathematical form of the claim that no further patch is possible. It should also be able to say why a principle is not just a sequence of local patches.

Poincaré's formulation and Einstein's both pass. Nothing observable separated them in 1905, and scoring Einstein higher would be the Whig framing of Section 3 applied as a grading rule. The advantage of Einstein's route appeared only later, at the gravity stage. Once spacetime is treated as geometry, the choice of field rank disappears. Keeping the aether is therefore not an error in this test, but a branch whose cost had not yet come due. One thing does earn extra credit: noticing that the two formulations are observationally indistinguishable on the evidence available. This is the capability listed against task 5 in Table 1.

5.3. The Gravity Test

The third test fixes the setting and asks four questions. Spacetime is flat and Minkowskian. Matter carries a symmetric second-rank energy-momentum tensor $T_{\mu\nu}$ obeying $\partial^\mu T_{\mu\nu} = 0$. The gravitational field is a field on this flat spacetime, coupled to matter, and Newtonian gravity has to be recovered in the static weak-field limit. The system is asked to enumerate the possible field types and say on what grounds; to write the coupling and the field equations for each; to compute for each the Newtonian limit, the perihelion precession of Mercury, the deflection of starlight grazing the Sun, and the gravitational redshift; and to say which candidates can be eliminated on paper, which can be separated only by observation, and what that observation is. The equivalence principle, Riemannian geometry, Einstein and the 1919 measurement are not mentioned.

Two kinds of answer fail. Producing a single candidate fails, whether it is the scalar that continues Poisson's equation or the tensor that happens to be right, because the space was never generated. Computing the observables of the vector theory fails, since a one-line argument has already disposed of it.

Success is the content of Table 2. The candidates are the scalar, the vector and the second-rank tensor, and the answer should say why the enumeration is not pushed higher rather than claim that it is provably complete. The vector goes with no computation at all, because a vector field makes like sources repel. The system should also rank the three measurements. Redshift is correct in both survivors and decides nothing. Mercury's target can be moved, since solar oblateness or an unseen mass can absorb part of the $43''$ (Dicke & Goldenberg, 1967), and the data predate the theories in any case. Light deflection is exactly zero in the scalar theory for a structural reason, cannot be tuned away, and had not been measured.

What is scored is the route rather than the answer. The Einstein equations can be reached from this starting point without ever leaving the flat background, by coupling a rank-two field to $T_{\mu\nu}$, then to its own energy, and summing the series, as Gupta, Kraichnan, Feynman and Deser each showed between 1954 and 1970 (Gupta, 1954; Kraichnan, 1955; Feynman et al., 1995; Deser, 1970). That answer is within the premises. The same equations justified by the curvature of spacetime are not, since the setting fixes the field as living on a flat background. The two are distinguishable only if the derivation is demanded along with the result.

6. Discussion and Conclusion

- Restate: the question is not whether a system reaches Einstein's answer.
- The decisive test is whether it can reconstruct the space of possibilities, recognise what remains undecidable on paper, and determine which question must next be put to nature.
- Where the division of labour falls: enumeration and paper-elimination on one side, adjudication on the other.
- A system working in mathematics can in principle certify its own result, and a system working in physics cannot; Section B sets out the contrast.
- Limitations: a single case; the enumeration in Section 5.3 uses representation-theoretic tools unavailable before 1939; the tests are conceptual.

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A. The Three Candidate Theories

- Couplings, field equations, and derivations of the four observables.
- Timeline of experiments and theories, 1859–1919.

B. Mathematics Closes Its Loop, Physics Does Not

A system doing mathematics can certify its own output. A formal proof can be checked step by step. The checker is software, and correctness is decidable relative to the axioms that were assumed. Generation and verification therefore live in the same box. That is what makes the formal-mathematics programme discussed in Section 3 coherent as an engineering goal (Hubert et al., 2025): a system can propose a proof, run the checker, and know the answer without consulting anything outside itself.

Physics has no such checker. A theory can be tested internally for Lorentz covariance, conservation, stability, and recovery of the known limit, and Nordström's theory passes all of these tests. It is self-consistent, covariant, and attractive. It explicitly requires $m_i = m_g$, and it gives the correct gravitational redshift. Every check that can be run at the desk returns a pass. It is also wrong. No further internal scrutiny separates it from the tensor theory, because the two disagree in only one place, and that place is a number that has to be measured.

Simulation does not close the gap. A simulation runs the theory under test, so it returns what the theory already contains. It cannot supply an independent verdict. Automating the apparatus does not close the gap either. Execution can be handed to self-driving laboratories and robotic observatories (Boiko et al., 2023), and nothing in the argument concerns who operates the instrument. The claim is about where the information that settles the question comes from. It has to come from outside the theory being tested.

The dependence is not confined to a final validation step. The discriminating measurement usually does not exist until a theory calls for it. The experiments of Rayleigh, Brace, Morley–Miller and Trouton–Noble were designed to test consequences that Lorentz's contraction hypothesis had generated. The 1919 eclipse expedition was mounted because a theory had named a specific number. Theory and experiment are interleaved rather than sequential. That is why the loop cannot be closed once at the end, and why a human, or at minimum the world, sits inside it rather than after it.

One qualification matters. The argument concerns the discovery of theories, not computation carried out under a theory that is not in question. When the governing physics is settled and the difficulty is scale, the verification loop is much closer to being closable inside the machine. Examples include predicting a material property from established quantum mechanics (Merchant et al., 2023) or a protein structure from established biochemistry (Jumper et al., 2021). Results of that kind do not contradict anything here. The asymmetry appears exactly when the theory itself is what is at stake.

The practical consequence for evaluation is that the deliverable from a system doing theoretical physics is not just a theory. It is a theory together with the observation that would kill it. Scoring the pair is what the three tests in Section 5 are designed to do.